

Safety-based path finding in urban areas for older drivers and bicyclists



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ARTICLE INFO

Article history:

Received 17 March 2014

Received in revised form 24 August 2014

Accepted 24 August 2014

Available online 18 September 2014

Keywords:

Older drivers

Bicycle path

Safe path

Safety indicator

Multi-objective function

ABSTRACT

This paper presents a safety-based path finding methodology for older drivers and bicyclists in an urban area. The paths are estimated based on costs consisting of both safety and travel time. Safety is evaluated against potential risk of a crash involving an older driver (or a bicyclist) with other vehicles present on the road. To accomplish this, simple formulations are developed for safety indicators of streets and intersections, which are actually generic irrespective of the type of road user. Traffic attributes such as speed and density, driver attributes such as perception–reaction time and street attributes of length and tire-to-road friction coefficient are taken into account in building the safety indicators. Thus, the safety indicators do not necessarily require historical crash data which may or may not be available during path finding. Subsequently, a multi-objective shortest path algorithm is presented that identifies the best path (the non-inferior path) from amongst a set of selected safest paths with due considerations to travel time incurred on each. A simple application example of the proposed methodology is demonstrated on an existing street network system from the City of College Station, Texas. The contributions of this research are twofold – first, the safety indicators can be used by planners in determining high crash potential sites – streets and/or intersections – and second, the safety-based path finding methodology developed in this paper can be integrated with modern day route planning devices and tools in guiding older drivers and bicyclists within an Intelligent Transportation Systems framework.

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1. Introduction and background

Traffic safety has been a growing concern worldwide since an estimated 1.2 million people are killed in road crashes annually and as many as 50 million are injured (WHO, 2013). As also estimated by the World Health Organization (WHO), with the current trend, the percentage of these crashes would increase by about 65% over the next 20 years. In fact, traffic crashes are one of the leading causes of deaths in the United States. Ranked 5th in terms of the years of life lost (i.e., the number of remaining years that the person is expected to live had they not died), crash-related deaths are behind only cancer and heart diseases (Subramanian, 2012). Among different types of drivers on road, a high number of nonfatal as well as fatal crashes are observed that involve older drivers (Classen et al., 2010) and bicyclists (Kerr et al., 2013). Due to age-related weaknesses, there is an increased crash risk for older drivers (Awadzi et al., 2008). In case of bicyclists, as per the recent data released by the National Highway Traffic Safety Administration (NHTSA) in the United States, there is an esti-

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mated 49,000 cyclists who were injured and as high as 726 of them were killed in motor-traffic related crashes in 2012 (NHTSA, 2014). Thus, safety-based mobility for both motorized (especially, for older drivers) and non-motorized travelers (such as for bicyclists) should be appropriately addressed and analyzed.

With an aim to reduce traffic crashes, policies that influence decision-making in designing and modeling routes from the point of view of Intelligent Transportation Systems (ITS) (Adler and Blue, 1998) and also within the transportation planning perspectives (Litman, 2012) have become very important in recent times. Related to the use of ITS in safety, in 2006–2009, the USDOT and the Crash Avoidance Metrics Partnership-Vehicle Safety Communications 2 (CAMP-VSC2) Consortium (consisting of major automobile manufacturer giants in the United States) realized several limitations for the Autonomous Safety Systems in crash imminent scenarios. The limitations of the safety systems consisted of late confirmation of a stopped lead vehicle as an in-path stationary target, and occasional incorrect out-of-path target detection and incorrect vehicle rejection (Ahmed-Zaid et al., 2011). The root cause for these failures was identified as shortcomings from the algorithms that were used in the safety systems. These algorithms were reliable of detecting vehicles only at close target range and also had poor object and path predictions at certain curve to straight road transitions.

Cooperative collision warning devices are often required for vehicles navigating on streets. In recent times, collision avoidance features have been rapidly getting integrated into the modern day vehicle fleet. These features involve on radar or camera assisted technologies that cause vehicles to activate brakes prior to potential collisions. Most of these collision warning systems (CWS) function utilizing wireless communication technologies and laser-based systems such as radars and video sensors. Often during normal functioning, a CWS device becomes limited by the line-of-sight obstruction to detect and respond to vehicle or obstacle presence. Thus, researchers have developed tunable algorithms which only require knowledge of the vehicle's location, speed and its brake pressure for safe navigation with the aim to make CWS more efficient (Miller and Huang, 2002).

In order to gather information regarding location, speed and other relevant attributes of a vehicle in a network, a communication device such as a Global Positioning System (GPS) is often employed. However, a GPS device is not always reliable (or robust) for rendering a constant and uninterrupted communication between vehicles (Boukerche et al., 2008). There are inevitable failures associated with the functionality of GPS and these failures are often attributed to weak satellite signals that cause inaccuracies in the location of vehicles and sometimes the vehicles are not at all detected in a network (Basiri et al., 2012; May and LaPierre, 2008; Otto et al., 2009). Therefore, persistent outages of GPS and related ITS-based radar devices entail need for alternate solutions for path finding purposes which would not require sole dependency on devices prone to failures at critical moments (Kukshya et al., 2005).

Factors on which traffic safety depends vary with the type of travel activity, built environment, roadway and vehicle type, and driver behavior. However, the manner in which safety is impacted by these factors is still under investigation by the roadway designers. Dumbaugh and Gattis (2005), and Ewing and Dumbaugh (2009) conclude that traffic environments of dense urban areas are safer than those in the suburbs. Thus, reiterating that built environment has an impact on the crash frequency. This has subsequently helped in understanding several other missing elements affecting traffic safety and the findings entail further improvements in roadway design. However, with new improvements, a perfect safe environment for road is not often achieved at the very first shot. This is due to uncertainty and randomness involved in decision-making process of drivers who perform various complex traffic maneuvers.

There have been suggestions from safety experts in promoting ideas and strategies for improving intersection safety in communities (Porter and England, 2000). From the perspective of path finding, researchers have presented different types of cost-effective routes on a network, such as determining an optimal path based on the least cost travel time (Goczyła and Cielatowski, 1995) and other types of costs such as eco-friendly routes (Nie and Li, 2013). A route planning technique based on the concept of identifying important nodes in a network was presented by Geisberger et al. (2008). Similarly, Hood et al. (2011) estimated a GPS-based bicycle route model for San Francisco, California, in order to understand the decision-making choice of cyclists. Factors such as the effect of steep slopes, traffic volume, traffic speed etc. were investigated on route choice. Nevertheless, no exclusive analysis was performed for estimation of safety-based path for bicyclists or other types of road users. Dijkstra (2013) evaluated safety effects of a selected route choice in a regional network. However, the trade-off between safety-based route choice and the travel time involved on the selected best path was not analyzed nor was the diverse application of the technique underlined.

Intuitively, if safety becomes important in path finding, crash potential attributes of street links and intersections encountered on the path should be taken into account. This paper acknowledges this fact and develops analytical expressions for safety indicators which help rank streets and intersections of a given street network system. The safest path (via appropriate street links and intersections meeting required safety) is selected between a given origin and a destination point for two modes of transport – a passenger car (driven by an older driver) and a bicycle or a bike (operated by a cyclist).

There are several factors that govern safety of bicyclists on roads, such as bike-lane widths, presence of on-street parking, etc. (for details see Carter et al. (2006)). Our main focus, however, in this paper, is on presenting safety-based paths for a bike (as well as a passenger car) based on existing street speeds and traffic density.

We rank streets and intersections in a network based on their crash risk potential and propose a multi-objective function using both safety and travel time as costs to arrive at the best route termed as the 'non-inferior' path. From our exhaustive literature surveys and to the best of our knowledge, there is no single research up till date that evaluates safety performance of streets and intersections in a shortest path finding paradigm. This research also partially tries to overcome challenges and drawbacks associated with collision warning devices and GPS when the latter are crippled or non-existent.

For simplicity, from hereon, unless otherwise mentioned, we denote a passenger car (driven by an older driver) and a bike uniquely as ‘vehicle’ in order to maintain a crisp presentation of the idea on safety-based path-finding.

2. Safety considerations on a path

Safety is considered from the point of view of ‘crash risk potential’ of a vehicle (with other vehicles) as it navigates through streets and intersections. Crash risk potential is used as proxy to safety, particularly when there is absence of satisfactory crash data at a given site. Evaluating crash risk potential involves quantifying safety-related risks of traveling over a network by choosing the final best path (or, a non-inferior path) in terms of both safety and travel time. However, quantifying safety is quite complex due to different types of possible conflicts that occur on a street or at an intersection.

In the past, researchers have theoretically described crash or ‘collision potentials’ of vehicles traveling on two intersecting streets. Miller and Huang (2002) developed an algorithm for collision warning system of two-vehicles on route contention. It required coordinate locations of the two crash imminent vehicles to be known a priori through a GPS device in order to use the algorithm successfully.

Frequent crashes on a street occur when the front downstream vehicle suddenly brakes and stops without giving adequate warning or reaction time for the driver in the upstream rear vehicle to react and brake. This often results in rear-end collision. At intersections, there are conflicts related to stop sign violations, traffic signal violations, left-turn conflicts with approaching or crossing vehicles, and general crossing-path conflicts at uncontrolled intersections (Shladover and Tan, 2006).

In all collision imminent scenarios, whether on a street or at an intersection, the perception-reaction time of the driver to react and brake is important in deciding if a crash would occur. In order to properly examine the role of a driver in a crash, it’s essential to study the context or the built environment where the crash occurs. Considering human behavior and traffic attributes such as speed and density to be random, our first step consists of estimating ‘time-to-collision’ between any two vehicles traveling on street (call it s) and/or crossing an intersection (call it m). We use variables $t_{s,\tau}$ and $t_{m,\tau}$ to denote the respective times-to-collision on street s and intersection m at a given time, say τ . The safety indicators for a street and an intersection are derived based upon $t_{s,\tau}$ and $t_{m,\tau}$ for any random vehicle on a given network.

2.1. Time-to-collision on a street link

The crash risks on a street result from a variety of obstructions, such as presence of unexpected objects on the streets, interference from random jay-walkers/pedestrians, or from unexpected braking of a front vehicle. While there could be several other types of interferences that might be present on a street, which would potentially jeopardize traveling with safety, in this paper we focus on analyzing obstructions resulting solely from braking of a vehicle on a road under two different traffic density scenarios – sparse and dense. Therefore, the times-to-collision are redefined as $t_{s,\tau} = t_{q,s,\tau}$ under sparse (q) and $t_{s,\tau} = t_{Q,s,\tau}$ for dense (Q) traffic conditions of the road.

2.1.1. Sparse traffic conditions

In order to derive the expression for the time-to-collision of a vehicle with another vehicle sharing the same street, we assume a uniform “spatial” random distribution of vehicles on the street – as with this assumption the subsequent derivations of formulas become mathematically tractable as will be seen later. From the concepts of probability theory, the spatially uniform distribution of vehicles on a road follows a Poisson process (Larson and Odoni, 1981). Thus, the probability $P(\Phi = b)$ that there are b numbers of uniform spatially distributed vehicles on a street link of a network is given by the Poisson distribution as,

$$P(\Phi = b) = \frac{(\sigma_\tau^s g)^b e^{-\sigma_\tau^s g}}{b!} \quad (1)$$

where, Φ = random number of vehicles present on a street s , $b = 0, 1, 2, 3, \dots$, and g = distance of separation from a subject vehicle and the closest vehicle downstream on the same lane and direction as the subject vehicle (see Fig. 1). Parameter σ_τ^s is the density of vehicle defined as the number of vehicles per lane unit length of street s observed at time τ .

The probability that there is no vehicle at least distance g in front of the subject vehicle (see Fig. 1) can be estimated by substituting $b = 0$ in Eq. (1); in which case, the probability that the closest vehicle is present at distance g from the subject vehicle becomes $P(\Phi = 0) = e^{-\sigma_\tau^s g}$. Therefore, the expected closest distance (δ_q) between the subject vehicle and the closest vehicle traveling in front is given by,

$$\delta_q = \int_{g=0}^{L_s} P(\Phi = 0) dg = \int_{g=0}^{L_s} e^{-\sigma_\tau^s g} dg = \frac{1}{\sigma_\tau^s} (1 - e^{-\sigma_\tau^s L_s}) \quad (2)$$

where, L_s is the length of street s .

The expected time-to-collision ($t_{q,s,\tau}$) of the subject vehicle with the closest downstream vehicle in front is,

$$t_{q,s,\tau} = \frac{\delta_q}{S_s - S_0} \quad (3)$$

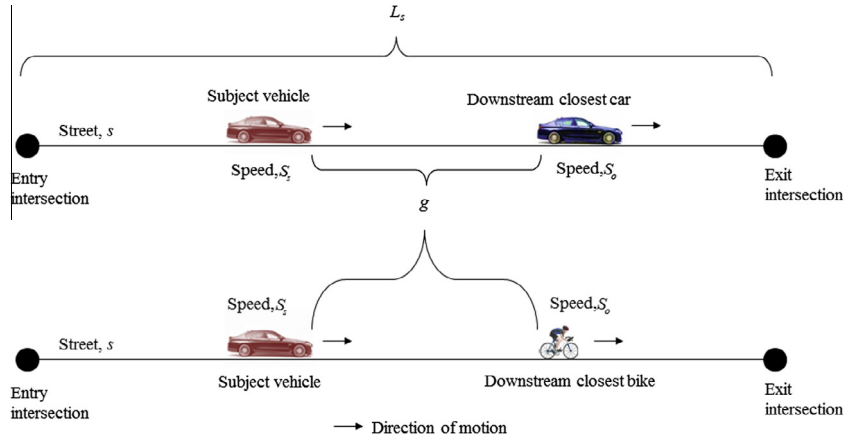


Fig. 1. Modeling spatial distance between two closest vehicles on a street.

where, variables S_s and S_o are the respective speeds of the subject vehicle and the closest downstream vehicle. The value of δ_q can be obtained from Eq. (2). The term $(S_s - S_o)$ in Eq. (3) is the relative sum of speeds for two vehicles traveling on the same lane of the street in the same direction. For simplicity, the speed S_s (or, S_o) can be assumed to be the average speed of the vehicle on street s . Speed S_s can also be estimated from historical speed data existing on the street, with S_o considered to be zero when the front vehicle brakes and stops suddenly.

2.1.2. Dense traffic conditions

Under congested traffic conditions, the instantaneous gap between successive vehicles is assumed to follow Gaussian Unitary Ensemble (GUE) distribution (Abul-Magd, 2007). The GUE distribution, $f(\delta)$, is expressed as,

$$f(\delta) = \frac{32\delta^2(\sigma_\tau^s)^3}{\pi^2} e^{-\frac{4\delta^2(\sigma_\tau^s)^2}{\pi}} \quad (4)$$

where, δ is the random spacing between two vehicles on a street, and σ_τ^s is the density of vehicles on street s evaluated at time τ . The expected closest distance (δ_Q) between the subject vehicle and the closest vehicle traveling downstream is given by,

$$\delta_Q = \int_0^{L_s} \delta f(\delta) d\delta = \int_0^{L_s} \frac{32\delta^3(\sigma_\tau^s)^3}{\pi^2} e^{-\frac{4\delta^2(\sigma_\tau^s)^2}{\pi}} d\delta = \frac{1}{\sigma_\tau^s} \left(1 - \frac{4L_s^2(\sigma_\tau^s)^2}{\pi} e^{-\frac{4L_s^2(\sigma_\tau^s)^2}{\pi}} - e^{-\frac{4L_s^2(\sigma_\tau^s)^2}{\pi}} \right) \quad (5)$$

Therefore, using S_s and S_o as the respective speeds of the subject vehicle and the downstream closest vehicle, the expected time-to-collision ($t_{Q,s,\tau}$) between the two is,

$$t_{Q,s,\tau} = \frac{\delta_Q}{S_s - S_o} \quad (6)$$

where the value of δ_Q can be obtained from Eq. (5).

2.2. Time-to-collision at an intersection

Research shows that a high number of traffic crashes occur at intersections (Ferreira and Couto, 2013). Consider two vehicles A and B that are in route contention (see Fig. 2). Suppose, the time-to-collision of subject vehicle A with another vehicle B at the intersection (say, m) is $t_{m,\tau}$, (where $m \in M$, number of intersections in the network) at a given time τ . Intuitively, $t_{m,\tau}$ should be sensitive to the density of the vehicles present around the intersection, i.e. the number of vehicles per unit street length and the speeds of two closest approaching vehicles that are in route contention. Further, assume intersection m has n -legs (see Fig. 2). Another vehicle B is approaching the same intersection m and is the closest to vehicle A than any other vehicle located on any of the $(n - 1)$ remaining legs. In the sketch of Fig. 2, vehicle A is traveling on the s th leg of the intersection with an approach speed S_s^m towards the intersection m . Assume that the other closest vehicle B approaching the same intersection on an adjacent leg j has a speed of $S_{s,j}^m$, with j being one of $\{1, 2, \dots, n\}$ and $s \neq j$. Thus, the expected closest distance between subject vehicle A and the closest vehicle B approaching the common intersection m can be derived. We denote by Ω_q this expected closest distance between the two vehicles A and B for sparse traffic conditions and Ω_Q for dense or congested traffic conditions.

The expected value of time-to-collision $t_{m,\tau}$ is obtained by the ratio of Ω_q (or, Ω_Q) over the relative speeds, $(S_s^m + S_{s,j}^m)$, of the two vehicles approaching the intersection m .

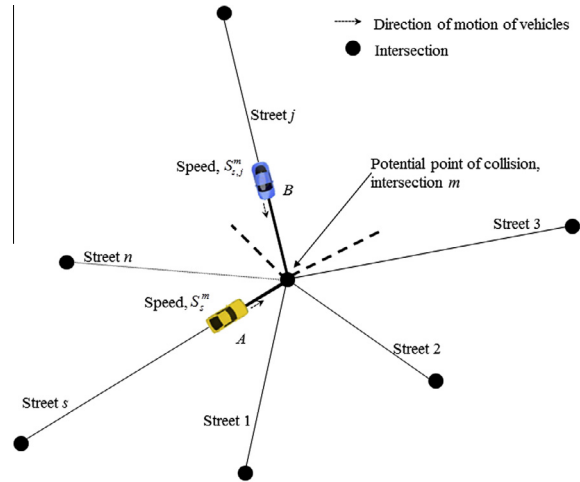


Fig. 2. Vehicle collision modeling at an intersection.

2.2.1. Sparse traffic conditions

Similar to the assumption made earlier for the spatial distribution of vehicles on a street, we assume that vehicles around an intersection are uniformly and randomly distributed. Therefore, random street-based distance between any two vehicles closest to the intersection follows a Poisson distribution. The probability of having v number of vehicles present around the intersection is $P(V = v)$ and is expressed as,

$$P(V = v) = \frac{(\rho_\tau^m D)^v e^{-\rho_\tau^m D}}{v!} \quad (7)$$

where, V = random number of vehicles present around an intersection m with $v = 0, 1, 2, 3, \dots$. Variable D = sum of the lengths of the streets converging at the intersection m and ρ_τ^m is the density of vehicles (defined as number of vehicles per unit length) around the intersection m at time τ .

The probability that vehicle A does not encounter a vehicle up to at least total distance D around the intersection m should be such that v is equal to 0, in which case, $P(V = 0) = e^{-\rho_\tau^m D}$. Therefore, the expected closest distance, Ω_q (for sparse traffic condition) between any two vehicles around the intersection m is then,

$$\Omega_q = \int_{D=0}^{D=D_{\max}} P(V = 0) dD = \int_{D=0}^{D=D_{\max}} e^{-\rho_\tau^m D} dD = \frac{1}{\rho_\tau^m} (1 - e^{-\rho_\tau^m D_{\max}}) \quad (8)$$

Further simplification of (8) involves estimating the value of $D_{\max}$. Estimating $D_{\max}$ is important in truly estimating the expected gap between two approaching vehicles. In reality, it is very likely that the density ρ_τ^m around an intersection would be approximately equal in value to the density observed on the street merging to the intersection m , especially either in extreme sparse or extreme dense traffic road conditions. Thus, for such situations, the vehicular density on street s (if merging onto intersection m), denoted earlier in the paper as σ_s^m , will be close to ρ_τ^m . With this argument, $D_{\max}$ can be approximately equated to summation of all of the street legs that converge at the intersection m , i.e. $D_{\max} = \sum_{u=1}^n L_u$, where, L_u = length of street segment u converging at the intersection with $u = \{1, 2, \dots, n\}$. In which case (8) becomes,

$$\Omega_q = \frac{1 - e^{-\rho_\tau^m (L_1 + L_2 + \dots + L_n)}}{\rho_\tau^m} \quad (9)$$

The expected travel time, (i.e. say, $t_{m,\tau} = t_{q,m,\tau,s}$), to collision for any vehicle traveling on street s towards an intersection m under sparse traffic condition is,

$$t_{q,m,\tau,s} = \frac{\Omega_q}{S_s^m + S_{s,j}^m} \quad (10)$$

where, the value Ω_q can be estimated from (9). The speed of the subject vehicle approaching the intersection m is S_s^m and $S_{s,j}^m$ is the speed of the vehicle on one of the adjacent incoming links j to the same intersection m .

2.2.2. Dense traffic conditions

We assume the GUE distribution (see Eq. (4)) to estimate the expected closest distance between two vehicles approaching an intersection under 'congested' traffic conditions. Taking cue from the expression in Eq. (4), the expected distance between any two vehicles around an intersection m can be derived. This can be achieved by replacing L_s with sum of all the lengths ($\sum_{u=1}^n L_u$) of the n streets incoming to the intersection m . Therefore, the expected distance (Ω_Q) between any two vehicle closest to the intersection m at time τ is given by,

$$\Omega_Q = \frac{1}{\rho_\tau^m} \left(1 - \frac{4(\sum_{u=1}^n L_u)^2 (\rho_\tau^m)^2}{\pi} e^{-\frac{4}{\pi} \left(\sum_{u=1}^n L_u \right)^2 (\rho_\tau^m)^2} - e^{-\frac{4}{\pi} \left(\sum_{u=1}^n L_u \right)^2 (\rho_\tau^m)^2} \right) \quad (11)$$

The expression for the expected travel time for dense traffic condition, say $t_{m,\tau} = t_{Q,m,\tau,s}$ can be obtained as,

$$t_{Q,m,\tau,s} = \frac{\Omega_Q}{S_s^m + S_{s,j}^m} \quad (12)$$

In the above expression of Eq. (12), the value for Ω_Q can be obtained from Eq. (11).

3. Critical time-to-avoidance

The ‘critical’ time-to-avoidance is defined as the minimum time of separation between any two moving vehicles on a street so that collision can be averted when brakes are applied. We express the critical time-to-avoidance of a vehicle with another vehicle with λ_{crit} (say λ_{crit}^s on a street s or $\lambda_{crit}^{m,s}$ for an intersection m with vehicle being on street s incoming to intersection m). Some of the main factors on which λ_{crit} depends on are the agility of the vehicle, the response time of the driver, and the time characteristic of the collision avoidance measure chosen (Miller and Huang, 2002). Not all of these factors can be quantified easily to be used for developing the λ_{crit} . However, a basic ingredient that should be used as an indicator of λ_{crit} is the response time, r , of the driver. The expression proposed by Miller and Huang (2002) for λ_{crit} is used in this paper and is expressed as,

$$\lambda_{crit} = \left(r + \frac{\beta S}{\mu a} \right) \quad (13)$$

where, μ is the estimated vehicle tire-to-road friction coefficient, a is the acceleration due to gravity, S is the speed of the vehicle on the street s (i.e. $S = S_s$), or S could also be approach speed of a vehicle to intersection m (i.e. $S = S_s^m$). Parameter β is the speed reduction factor and is a non-negative number. A value of $\beta = 0$ would mean an abrupt vehicle stop.

The coefficient of friction μ is between the road and the tires of the vehicle and impacts the stopping distance (or, time) of the vehicle. However, the value of μ has ambiguities which could impact λ_{crit} . For example, on dry road conditions, it's comparatively easier to halt the vehicle by applying brakes than compared to on wet condition on the roads. Therefore, for dry conditions, a high value of μ is expected as compared to that for a wet road. It has been observed that there is an increase in accident rates in the rainy season due to very low values of μ (Asi, 2007). Therefore, using a suitable value of the friction coefficients in estimating λ_{crit} is critical in fine tuning the safety indicators derived in the next section.

4. Safety indicator formulation

Constraints can be developed with the help of critical time-to-avoidance, λ_{crit} , for proposing safety indicators for streets and intersections. Suppose, the critical time to avoidance on a street link s is expressed as $\lambda_{crit} = \lambda_{crit}^s$. Therefore, traveling on a street is safer whenever $\frac{\lambda_{crit}^s}{t_c^s} < 1$ and relatively unsafe when $\frac{\lambda_{crit}^s}{t_c^s} \geq 1$. For a vehicle traveling on street s towards an intersection m , the critical time-to-avoidance, say $\lambda_{crit} = \lambda_{crit}^{m,s}$, is such that for the inequality $\frac{\lambda_{crit}^{m,s}}{t_{Q,m,\tau,s}} \geq 1$ intersection m is relatively unsafe to pass through as compared to the situations when $\frac{\lambda_{crit}^{m,s}}{t_{Q,m,\tau,s}} < 1$. Thus, the non-inferior path that we are seeking should take into account the above constraints as precursors of safety on streets and at intersections.

We propose two indicators of safety in order to quantify crash potential of a vehicle as it travels on a street network. The first indicator is the safety indicator for a street link s (i.e. I_τ^s), quantifies how safe two vehicles are while traveling on a common street s at time τ . The second indicator (i.e. $I_\tau^{m,s}$) is evaluated for an intersection m at time τ for the subject vehicle traveling on an incoming link s to the intersection. Both of these indicators, I_τ^s and $I_\tau^{m,s}$, are based on the critical time-to-avoidance.

4.1. Safety indicator – street

To summarize, with the expansion of terms $t_{q,s,\tau}$ and $t_{Q,s,\tau}$ using Eq. (2), and Eq. (5), respectively, for any link s , the final expression for I_τ^s is

$$I_\tau^s = \begin{cases} \frac{\lambda_{crit}^s}{t_{q,s,\tau}} = \frac{\lambda_{crit}^s (S_s - S_0)}{\delta_q} = \frac{\sigma_\tau^2 (r + \frac{\beta S_s}{\mu a}) (S_s - S_0)}{(1 - e^{-\sigma_\tau^2 L_s})} & \text{for sparse traffic} \\ \frac{\lambda_{crit}^s}{t_{Q,s,\tau}} = \frac{\lambda_{crit}^s (S_s - S_0)}{\delta_Q} = \frac{\sigma_\tau^2 (r + \frac{\beta S_s}{\mu a}) (S_s - S_0)}{\left(1 - \frac{4L_s^2 (\sigma_\tau^s)^2}{\pi} e^{-\frac{4L_s^2 (\sigma_\tau^s)^2}{\pi}} - e^{-\frac{4L_s^2 (\sigma_\tau^s)^2}{\pi}} \right)} & \text{for dense traffic} \end{cases} \quad (14)$$

4.2. Safety indicator – intersection

For an intersection m , having n converging links/legs with two closest incoming vehicles traveling on street link s and j ($s \neq j$), the safety indicator, $I_{\tau}^{m,s}$, can be written as,

$$I_{\tau}^{m,s} = \begin{cases} \frac{\lambda_{crit}^m}{t_{q,m,\tau,s}} = \frac{\lambda_{crit}^m (S_s^m + S_j^m)}{\Omega_q} = \frac{\rho_{\tau}^m \left(r + \frac{\beta S_s^m}{\mu a} \right) (S_s^m + S_j^m)}{\left(1 - e^{-\rho_{\tau}^m \left(\sum_{u=1}^n L_u \right)} \right)} & \text{for sparse traffic} \\ \frac{\lambda_{crit}^m}{t_{Q,m,\tau,s}} = \frac{\lambda_{crit}^m (S_s^m + S_j^m)}{\Omega_Q} = \frac{\rho_{\tau}^m \left(r + \frac{\beta S_s^m}{\mu a} \right) (S_s^m + S_j^m)}{\left(1 - \frac{4 \left(\sum_{u=1}^n L_u \right)^2 (\rho_{\tau}^m)^2}{\pi} e^{-\frac{\pi}{2} \left(\sum_{u=1}^n L_u \right)^2 (\rho_{\tau}^m)^2} - e^{-\frac{\pi}{2} \left(\sum_{u=1}^n L_u \right)^2 (\rho_{\tau}^m)^2} \right)} & \text{for dense traffic} \end{cases} \quad (15)$$

The expressions from Eqs. (9) and (12) are used for expanding $t_{q,m,\tau}$ and $t_{Q,m,\tau}$, respectively, in Eq. (15). Note that lower the value of the safety indicator shown in Eqs. (14) and (15), the lesser the crash risk is at the respective street (or at the selected intersection). Therefore, every street link and intersection of a network can be ranked with the indicators shown in Eqs. (14) and (15).

For a path selection solely based on safety considerations, the safety-based ranking with indicators using Eqs. (14) and (15) can be used. The aim will be to avoid those street links and intersections that have high safety indicator values encountered while traveling from a given origin intersection to a known intersection destination.

4.3. Validation using observations from safety research

The indicators presented in Eqs. (14) and (15) are directly proportional to the vehicular density, speed, and reaction time of the drivers. The reaction time of a driver is directly proportional to the minimum stopping distance (Son et al., 2011). Thus, greater the reaction time, higher is the minimum stopping sight distance and hence, more likely the chance of a crash and vice versa. The safety indicators in Eqs. (14) and (15) are in agreement with this argument.

Further, it has been found that increase in speeds leads to higher accident rates (Aarts and van Schagen, 2006). Safety indicator values would increase when vehicular speeds (in the numerators of Eqs. (14) and (15)) are increased, thereby increasing the potential of a crash.

Although, there have not been many literatures describing the relationship between vehicular density and accidents (Wang et al., 2013), observations from crash data on urban freeway segments show that an increase in crash rates occurs with increase in density within certain limits (Lord et al., 2005). It is noted from these literatures that the 'crash rate potential' does increase in some manner with increase in traffic density. This is analogous to increase in safety indicator values of Eqs. (14) and (15) with increase in traffic density.

Thus, the discussion above serves to validate the safety indicators (in Eqs. (14) and (15)) as a good proxy for crash risk potential on a street or at an intersection. We proceed to show the application of these indicators, albeit for static case when time τ is invariable, in a safety-based path finding process in the next section.

5. Safety-based path finding: an application framework

For a relatively large network with several street links and intersections, it is often computationally difficult to find an optimal path which is simultaneously the safest as well as the shortest in travel time. The problem of finding a set of paths that minimizes two or more objective functions (such as maximizing safety and minimizing travel time) is referred to as multi-objective shortest path problem (Iori et al., 2010) and is known to be NP-complete (Michael and Johnson, 1979). Recently Pahlavani and Delavar (2014) studied a multi-criteria route planning based on a driver's preferences in multi-criteria route selection. In this paper, we propose a modified form of the multi-objective shortest path (MSP) algorithm based on median shortest path problem (originally proposed by Current et al. (1987)) to suit our needs. Essentially, we aim to find the non-inferior solution for the MSP.

5.1. MSP problem formulation

Consider a strongly connected directed network $G = (M, A)$ where $M = \{1, \dots, m\}$ is a set of intersections and A (with a street link $s \in A$) is a finite set of links from among the intersections from M . Thus, the objective for the MSP algorithm would be to find the shortest path from a given source intersection $m_s \in M$ to a destination intersection $m_d \in M$, simultaneously considering safety and travel time. The following formulations are presented for the MSP.

$$Z_1^p = \sum_i \sum_g (I_{\tau}^{s \rightarrow (i,g)} + I_{\tau}^{s \rightarrow (i,g)} X_{s \rightarrow (i,g)}^p) \quad (16)$$

$$Z_2^p = \sum_{h=1}^{R_p-1} (\Gamma_{s \rightarrow (h,h+1)}^p) \quad (17)$$

where,

- $s \rightarrow (i, g)$ is the street s with the beginning intersection i and ending intersection g
- $R_p = \{h|h \text{ is an intersection on the } p\text{th safest path}\}$
- $\Gamma_{s \rightarrow (h, h+1)}^p$ = travel time on street s that is on the p th safest path. Intersection h is on the p th safest path.
- $X_{s \rightarrow (i, g)}^p = \begin{cases} 1, & \text{if } s \text{ is on the } p\text{th safest path} \\ 0, & \text{otherwise} \end{cases}$
- $I_{s \rightarrow (i, g)}^p =$ safety indicator for street s
- $J_{s \rightarrow (i, g)}^p =$ safety indicator for intersection g with the subject vehicle traveling on s and the closest vehicle on route contention on street j which is one of the converging legs of intersection g

Variable Z_1^p in Eq. (16) finds the p th-shortest paths solely based on safety. Function Z_2^p in Eq. (17) estimates the travel time on the p th path.

5.2. MSP solution algorithm

Following steps are introduced to obtain the non-inferior solution for the problem formulation in Section 5.1.

- **Step 1.** Set $p = 0$. Identify P -safest paths from the starting intersection (say, $m_s \in M$) to the terminal intersection (say, $m_D \in M$).
- **Step 2.** Set $p = p + 1$. Define Z_1^p, R_p
Calculate Z_2^p .
If $p = P$, Then Go to Step 3, Else Repeat Step 2.
- **Step 3.** Set $p = 1$. Add (Z_1^1, Z_2^1) to the master list.
- **Step 4.** Set $p = p + 1$.
If $Z_2^p < Z_2^l$ for $l = 1, \dots, P - 1$, Then Add (Z_1^p, Z_2^p) to the master list.
If $Z_1^p = Z_1^l$ and $Z_2^p < Z_2^l$ for any $l = 1, \dots, P - 1$, Then Remove (Z_1^l, Z_2^l) from the master list.
If $p = P$, Then Go to Step 5, Else Repeat Step 4.
- **Step 5.** STOP.

The master list contains the set of non-inferior solutions.

For Step 1, Yen's algorithm can be used to find the p -shortest path based on safety (Yen, 1971). The time complexity for Yen's algorithm is $O(pM(A + M \log M))$, where A and M are the number of links and intersections in the network, respectively.

5.3. Application example

Mathematical construct of the safety indicators in Eqs. (14) and (15) allows one to rank streets and intersections for any type of street network system. We use this flexibility provided by the indicators to demonstrate application of the proposed idea presented in this paper on an existing street network from the City of College Station, Texas, USA (see Fig. 3). Bryan–College Station area is a hub for more than 55,000 strong student population from Texas A&M University and nearby Blinn College in Bryan.

Besides being characterized by a distinct hierarchy of streets suitable for cars, College Station has several dedicated bike lanes and routes to promote bike ride for its residents (see Fig. 3). For the application example, we selected a section of the street network that has significant numbers of streets which are bicycle friendly and conducive for a bicyclist. The data for the existing bike routes were obtained from the City of College Station (City of College Station GIS Department, 2014).

Complementing the use of real street network, real-world traffic data for speeds, densities and lengths of the streets from College Station were also utilized for the application example. For simplicity, we assumed all street links within the network of Fig. 3 to be two-way. The purpose was mainly to show the application of the MSP algorithm for finding the non-inferior path. As a reminder, note that the non-inferior path is based on both safety and travel time considerations.

Two different kinds of transport modes with distinct driver types are selected for the demonstration – a passenger car driven by an older driver and a bicycle used by a younger adult. The goal of the application example is to show the effect of route choice for the two categories of road users – an older driver and a bicyclist – when safety becomes an important ingredient in path finding.

5.3.1. Set-up

The data for the lengths of the individual street links in the network shown in Fig. 3 were obtained from the ESRI shapefile available from the City of College Station (City of College Station GIS Department, 2014). There are a total of 263 intersections and 465 links in the network. A link here is defined as the part of the street between any two intersections.

The ESRI shapefile for the network of the City of College Station also provided the speed limits at different levels (A, B, and C) of street systems shown in Fig. 3. These street level nomenclatures were devised for convenience. Further, we generated random link speeds for cars in the range between ± 5 mph to the existing speed limits for the three street levels. The speed



Fig. 3. Street hierarchy and bicycle lanes from the street network section of College Station.

profile of the bike is assumed to be randomly varying between 8 and 10 mph over the individual dedicated bike links between intersections. The images in Fig. 4 show the speed and travel time distribution profiles on the street links for the cars in the network. Similar maps in Fig. 5 show the speeds and travel times on various links for the bike.

Travel time on a link as shown in Figs. 4 and 5 for the two modes, was computed by dividing street link length by link speed. The street link density was obtained using the observed traffic volume on the link and its speed. The densities around an intersection was assumed to be slightly higher (by 10–12 veh/mile/lane for each intersection) than those existing on the merging individual legs of the intersection for the practical assumption that car densities around intersections would be, on an average, greater than at any of its respective merging legs. However, with only negligible difference in density for an intersection and on its legs, a uniform density around an intersection could be assumed for successfully calculating the safety indicators. The image in Fig. 6 shows the final vehicular density distribution on the streets and around the intersections that were used for the application example. Through calculations, the speeds, densities and street lengths for each of the opposite lanes of a particular link in the network (as shown in Fig. 3) were found to be approximately equal. For accurate traffic density measurement on streets and around an intersection, density estimation algorithms can be used in which every vehicle location in a given network is tracked (Mao and Mao, 2013). However, this was out of scope of our work in this present research endeavor.

The origin and destination points (intersections) were selected to be identical for both the car and the bike (see Fig. 3). The non-inferior path from the origin to the destination is decided based on the level of safety and the travel time on the path. As also mentioned earlier, for the application example shown in this paper, the time τ at which paths were found is assumed to be invariable.

For both the car and the bike modes, the safety indicator values were calculated for streets using Eq. (14) and for the intersections using Eq. (15). In order to compute bike safety indicators from Eqs. (14) and (15), the car densities were used. In this paper, we considered density values below 50 veh/mile/lane for sparse and density above 50 veh/mile/lane for dense traffic conditions. The intersection safety indicator in Eq. (15) was calculated while traveling on street s and approaching intersection m . The obtained value was added to the safety indicator of street s . The map in Fig. 7 shows the final composite safety indicator values for street and intersection together to be used for Z_1^B of Eq. (16). Comparing Figs. 3 and 7, it is seen that for both the car and the bike mode, the safety indicator values of streets are high in majority for Level A and B streets



Fig. 4. Speed and travel time distribution for the car mode.



Fig. 5. Speed and travel time distribution for the bike mode.

characterized by high speeds and density values. Note that a lower value of safety indicator for a street (intersection) would mean a safer street (intersection) to travel on and vice versa. A lower value of safety indicator for bike lanes/routes as compared to car mode is observed mainly due to lower biking speeds in the range of 8–10 mph and also due to lower perception reaction time of 1 s of a bicyclist (as compared to 2 s for older driver). These perception reactions times are used at intersec-



Fig. 6. Traffic density distribution over the network.



Fig. 7. Safety indicator values for car and bike modes.

tions. For street safety indicators, the perception reaction time value used is of those drivers from other vehicles on the street and is assumed to be 1 s. The perception reaction time of an older driver was assumed to be a second higher than that of a bicyclist who is younger adult (Dozza, 2013).

Table 1

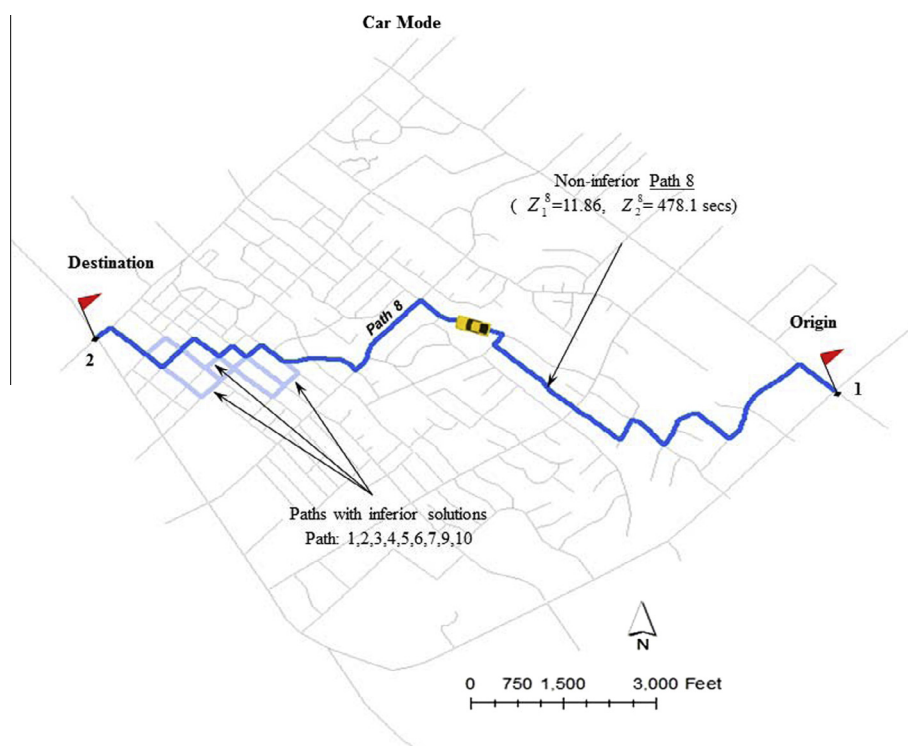
Parameter values (from Eqs. (14) and (15)) used for the application example.

SN.	Parameter/ variable	Definition	Value(s)
1	β	Speed reduction factor	0.001
2	μ	Vehicle tire-to-road friction coefficient	0.5
3	a	Acceleration due to gravity	22 mile/hour ²
4	S_o	Speed of car (or, bike)	Zero for car and non-zero for bike (refer Fig. 5 for bike speeds)
5	S_s	Speed of car traveling on street s	For car speed refer Fig. 4
7	S_s^m	Speed of car (or, bike) – traveling on street s towards intersection m	For car speed refer Fig. 4 and for bike speed refer Fig. 5
8	$S_{s,j}^m$	The speed of the car on an adjacent link j to street s and traveling towards intersection m	Refer Fig. 4
9	r	Driver/bicyclist perception reaction time	For the bicyclist, $r = 1$ s and for the older driver, $r = 2$ s (Dozza, 2013). For all other drivers $r = 1$ s
10	σ_τ^s	Traffic density on street s	Refer Fig. 6
11	ρ_τ^m	Traffic density around intersection m	Refer Fig. 6

Other assumptions and parameter values that are used in Eqs. (14) and (15), along with those mentioned above, are summarized under Table 1. Note that for the example presented, we ignore the time loss incurred at intersection while traveling. In situations when the numbers of intersections and street links in a network are extremely high, it might be computationally expensive to obtain a safety-based route. In such cases, a select size of the network can be analyzed in order to reduce computational times.

6. Results and discussions

We find the shortest paths for a value of $P = 10$. The paths are found using the MSP algorithm described in Section 5.2 and consist of both the non-inferior and the inferior paths. These paths are illustrated in Fig. 8 for the older driver and Fig. 9 for the bicyclist. The final selected non-inferior path is marked by a thick line, while rest of the inferior paths are represented using lighter colors. The MSP algorithm as described in Section 5.2 is applied separately for the paths of the car and the bike modes. All necessary computations were performed using programs developed in MATLAB R2013b.

**Fig. 8.** Safety-based paths for the car mode.

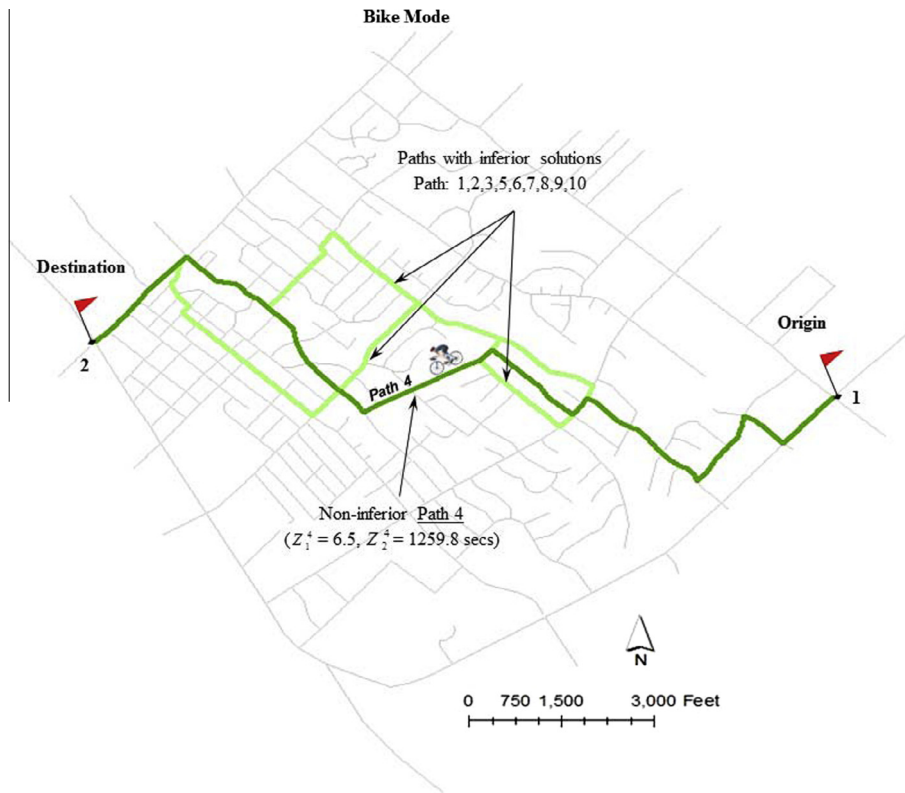


Fig. 9. Safety-based paths for the bike mode.

The values of path safety indicator, Z_1^p , are obtained for total of streets and intersections on the path with the safety indicator of each street and intersection computed using Eqs. (14) and (15), respectively. The values of Z_2^p are calculated for total of travel times on streets on a given path.

It is observed that in the quest of obtaining safest paths, all the ten paths utilize streets and intersections that have low safety indicator values (compare Fig. 7 with Figs. 8 and 9). From among the final ten paths obtained, the travel time range of the safety-based shortest path from the origin intersection (labeled 1) to destination intersection (labeled 2) for car varies between 478 and 499 s (see last column in Table 2). The range is 1259–1558 s for the bike mode (see last column in Table 3). As expected, the travel times by car are lower in value as compared to those by use of bike.

The total street safety achieved for the safe paths by car is in the range of 11 to 12 safety units (see second column in Table 2). The same for the bike in Table 3 are observed to be between 6 and 7 safety units. In general, we can also say that the path traveled by the bike appears to be comparatively safer than the car's. This can be mainly attributed to relatively lower operating speed of the bike as well as lower perception reaction time of the bicyclist as compared to that of an older

Table 2
Path attributes for the car.

Path (p) No.	Path safety indicator values Z_1^p	Path travel time Z_2^p (seconds)
1	11.71	499.1
2	11.77	495.4
3	11.77	497.7
4	11.80	497.8
5	11.83	494.0
6	11.85	496.7
7	11.85	494.1
8	11.86	478.0
9	11.89	496.2
10	11.89	496.6

Bold values denote non-inferior solution (i.e. non-inferior path attributes for the older driver).

Table 3
Path attributes for the bike.

Path (p) No.	Path safety indicator values Z_1^p	Path travel time Z_2^p (seconds)
1	6.01	1334.0
2	6.23	1288.0
3	6.47	1342.3
4	6.50	1259.8
5	6.61	1305.4
6	6.80	1429.5
7	7.02	1383.5
8	7.02	1558.4
9	7.04	1270.2
10	7.07	1313.7

Bold values denote non-inferior solution (i.e. non-inferior path attributes for the bicyclist).

driver. The results, however, can differ if other attributes/factors are taken into safety considerations for the bicyclist. This we leave for our future scope of study.

The non-inferior paths for the car and the bike mode have the travel times of 478 and 1259.8 s, respectively (see last column in [Tables 2 and 3](#)). The selected path which happens to be the non-inferior path is the one which has the minimum travel time, denoted by Z_2^p . Thus, for $p = 8$, the final safety-based path for the car mode has the value of $Z_2^8 = 478$ s and $Z_1^8 = 11.86$. For the bike mode with $p = 4$, $Z_2^4 = 1259.8$ s and $Z_1^4 = 6.5$. These values are marked in bold in [Tables 2 and 3](#).

7. Concluding remarks

Traffic safety is a growing concern worldwide, and particularly in the United States, traffic safety has been a concern in the urban areas which have high crash frequencies. Tools that promote safety-based path finding can be extremely useful in helping travelers with reduced vision and perception capacities (such as older drivers) and those that are at risk of an imminent collision or crash with other fast moving vehicles on roads (such as the bicyclists). For facilitating safe travel for these two types of road users, safety indicators for streets and intersections have been developed in this paper. The safety indicators are handy in ranking intersections and streets of a network. The lower the safety indicators for streets and intersections, the lower the risks are for a traveler of being involved in a crash. General observations from different safety-based traffic research validate the developed indicators proposed in this paper.

The safety-based indicators are integrated with the shortest path problem. With multiple paths available for a driver or a bicyclist in trying to reach a destination, the best path labeled as the 'non-inferior' path is found for maximal safety. The non-inferior path is the one which has the minimum travel time among the select safest paths identified.

In this paper, an example application of the safety-based path finding is shown over a section of real world street network system from the City of College Station, Texas. The safety-based non-inferior path on the College Station street network system had a cumulative safety indicator value of 11.86 and travel time of 478 s for a car driven by an older driver. For the bicyclist, the cumulative safety indicator of the non-inferior path was found to be 6.5 and travel time to be 1259.8 s. For both the modes, the origin and destination intersections were assumed to be identical. A lower value of safety indicator observed for the non-inferior path of the bicyclist was due to its lower operating speed of the bike and also due to lower perception reaction time of 1 s for the bicyclist. A higher value of perception reaction time was assumed for the older driver.

This research will have several far reaching contributions. The methodology outlined in this paper can be easily integrated with traditional Global Positioning System (GPS) devices to provide safety-based enhanced service to different types of transport users. In order to estimate a non-inferior path based on both safety and travel time, the inputs needed are the vehicular speeds and densities existing on a given street network system. These data can be normally obtained from history or even real-time traffic estimation devices. The attributes of a road such as street length and tire-to-road friction factor are by and large constant. Note, however, for evaluating these indicators, historical crash data (which might not be readily available while traveling) for a street or an intersection are not needed. Thus, the safety-based path finding methodology presented in this paper, though simplistic, is very useful for older drivers and bicyclists. This research is also useful as it provides an integrated safety-based and travel time-based application for assisting in vehicle navigation in critical areas of a street network system (such as city downtowns) where wireless and GPS devices are often known to malfunction. The safety-based path finding tool would also help in partially acting as backup against safety compromise due to failures or incapability of collision warning system devices.

Our future effort consists of reconstructing safety-based paths found in this research for older drivers and bicyclists incorporating other safety factors (not included in this study) for the two modes. Further, we also aim to develop real time safety-based shortest path algorithm and further streamlining its applicability and versatility for applications to transit systems and for pedestrians.

Acknowledgments

The author would like to thank Dr. Yunlong Zhang and Dr. Srinivas Geedipally for providing data for the City of College Station which helped fine-tune the application example used in this research. The author would also like to thank Dr. Luca Quadrifoglio for providing valuable resources for conducting this research. Finally, the inputs from anonymous reviewers are greatly appreciated which helped consolidate the final paper from its earlier submitted versions.

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